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A simulation based approach for enhancing automatic detection of cicada songs in challenging chorus conditions

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ABSTRACT

Advancements in machine learning-based audio recognition have significantly contributed to ecoacoustics. However, past studies have primarily focused on bird vocalizations, leaving insect sounds under-studied. One challenge in this area is recognizing insect calls in chorus conditions where multiple species vocalize simultaneously at high population densities. This complexity requires labeled training data specific to each chorus pattern, making sound recognition challenging and resource-intensive. To address this problem, we propose a data augmentation approach using physical acoustic simulation. Our method, tested on five species of Japanese cicadas, utilizes simulation software to generate numerous pseudo-chorus recordings by combining focal recordings of individual cicada vocalizations with background sounds. Thousands of simulations with various species combinations and individual counts were used to train a convolutional neural network (CNN) multi-label classifier. The proposed method achieved a 22% improvement in F1-score compared to models trained on focal recordings alone. By optimizing simulation parameters through hyperparameter tuning, we achieved a mean F1-score of 0.83 on real-world soundscape recordings with minimal training data (approximately 10 focal recordings per species). This approach is readily applicable to other chorusing taxa such as orthopterans and amphibians, offering a scalable solution for biodiversity monitoring in complex acoustic environments.

1. Introduction

Advancements in Passive Acoustic Monitoring (PAM) have been facilitated by the widespread availability of inexpensive Automated Recording Units (ARUs, e.g., AudioMoth (Hill et al., 2019)) and significant progress in machine learning methods for detecting animal vocalizations (Kahl et al., 2021; Ghani et al., 2023; Aodha et al., 2018). ARUs enable continuous and non-invasive acoustic data collection across various environments, providing unprecedented insights into biodiversity and ecosystem dynamics. This data accumulation facilitates tracking of species richness (Wimmer et al., 2013; Müller et al., 2023), behavior (Hart et al., 2015; Sumitani et al., 2021), and phenology (Kimura and Sota, 2023; Buxton et al., 2016), which are essential variables for biodiversity assessment and conservation. Ecoacoustic monitoring reveals the impacts of land-use change, human activity, and climate stressors on wildlife through the analysis of acoustic interactions (Pijanowski et al., 2011), serving as a valuable tool for studying elusive species and long-term ecological trends.

Insects such as cicadas and orthopterans play crucial roles in soundscapes, influencing other species (Hart et al., 2015) and even affecting human psychology (Tokue et al., 2022) through their continuous calls across a wide frequency range. Despite their ecological importance, PAM-based studies focusing on insects are relatively few compared to those on mammals, birds, and frogs (Sugai et al., 2019; Nieto-Mora et al., 2023). In this study, we considered the main challenge in insect vocalization research as the complex and overlapping nature of choruses where multiple species vocalize simultaneously at high densities. Bioacoustic databases such as Xeno-Canto (https://xeno-canto.org) have facilitated the training of bioacoustic recognition models and are utilized in most recent ecoacoustic studies. However, these databases primarily contain focal recordings targeting individual species' calls, which often result in decreased classification accuracy when applied to field recordings containing multiple overlapping calls and background noise (Kahl et al., 2020).

Among vocally active insects, cicadas are ecologically important organisms whose seasonal emergence patterns significantly impact ecosystems (Menninger et al., 2008). Their sustained, high-volume calls spanning a wide frequency range substantially influence soundscapes (Ross et al., 2018) and affect the vocal behavior of other species such as birds (Hart et al., 2015). Monitoring cicada species activity through acoustic methods provides valuable insights into their

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behavior, seasonality, and community interactions, particularly for species that are difficult to observe visually due to their arboreal habits. However, as discussed below, automatic recognition of complex cicada choruses has traditionally been a challenging task.

In this paper, we propose a novel approach that utilizes acoustic simulation to develop a deep learning model capable of accurately detecting 5 cicada species in eastern Japan in complex choruses. By artificially generating thousands of simulated choruses through combining individual cicada focal recordings and background sounds via physical acoustic simulation, we create a diverse training dataset that closely mimics real-world conditions. Our method enables accurate classification of multiple cicada species in chorus conditions with minimal training data (approximately 10 focal recordings per species). The model achieves a mean F1 score of 0.83 on real-world acoustic landscape recordings, representing a 22% improvement compared to models trained solely on focal recordings.

The main contributions of this study are: 1. Development of the first cicada-specific acoustic recognition model capable of identifying individual species within complex choruses 2. A novel data augmentation approach using physical acoustic simulation to generate pseudo-chorus recordings that mimic real-world acoustic environments, achieving high classification accuracy with minimal training data

2. Related work

2.1. Cicada song recognition

Research on automated cicada song recognition has primarily focused on identifying individual species from focal recordings. Tey et al. (2022) developed a convolutional neural network (CNN) model for cicada species classification but did not address recognition in chorus conditions where multiple species vocalize simultaneously. Similarly, Potamitis et al. (2007) demonstrated high accuracy in species identification of crickets and cicadas using probabilistic neural networks and Gaussian mixture models, but their approach was limited to focal recordings of individual species. Zilli et al. (2014) developed a hidden Markov Model-based acoustic detector for the New Forest Cicada as a smartphone application, achieving high accuracy in citizen surveys in the field, but it was restricted to a single species.

In contrast to these species recognition approaches, some studies have treated cicada choruses as noise to be filtered out from target sounds such as bird vocalizations. Zhang et al. (2024) developed CicadaNet, a deep learning model for filtering cicada choruses to improve long-term bird monitoring. Similarly, Brown et al. (2019) used multilayer perceptrons and support vector machines to filter rain and cicada noise from bird acoustic data. However, these approaches also do not attempt to identify individual cicada species present within choruses.

Research on other chorusing taxa, such as frogs, has also encountered challenges with reduced recognition accuracy due to chorusing. For example, Gan et al. (2020) investigated optimal data selection methods to improve frog call recognition performance in complex acoustic environments and reported that differences in chorus intensity between training and testing datasets adversely affected model performance. Additionally, Noda et al. (2016) reported that calls from non-target frogs and insects significantly degraded detection accuracy in developing their anuran call detection model.

There is a notable absence in the existing literature of research specifically focused on recognizing individual cicada species in complex chorus environments with multiple species, and our study aims to fill this important academic gap. Furthermore, the workflow we propose establishes a methodological foundation that can be applied to the study of other taxa that form vocal choruses by addressing the challenge of species identification in chorus conditions.

2.2. Data augmentation in bioacoustics

Data augmentation techniques have become essential tools for improving the performance of bioacoustic classification systems, particularly when training data is limited or when dealing with noisy recording environments. These techniques can be categorized into several approaches, each with specific applications and limitations.

Basic spectrogram augmentation techniques modify existing recordings through transformations such as time stretching, pitch shifting, and masking. SpecAugment, which applies time and frequency masking to spectrograms, has been shown to improve recognition accuracy for primate vocalizations (Ruan et al., 2022) and animal sounds in tropical rainforests (Sun et al., 2022).

Another approach involves mixing multiple recordings to create new training samples. MixUp, which linearly combines two different audio samples and their labels, has proven effective for bird sound recognition in complex acoustic environments (Kahl et al., 2020) and Anuran call classification (Moummad et al., 2024). However, these methods typically combine only two or a few sound sources and do not fully represent the complexity of natural choruses involving dozens of individuals from multiple species.

Noise-related techniques add or modify environmental noise in recordings. Adding background noise has been effective in improving classification in noisy environments (MacIsaac et al., 2024). These approaches enhance model robustness to noise but do not adequately address the specific challenges of chorus conditions with overlapping calls of multiple species.

Advanced generative methods create entirely new audio samples. Deep generative models such as Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs) have been used to generate synthetic bioacoustic data, with Herbst et al. (2024) comparing their effectiveness in bioacoustic classification. Physics-based simulations that consider the physical properties of sound propagation have been applied to cyber–physical systems (Luo et al., 2021) and sound event localization (Wang et al., 2023).

Our approach falls into this last category, employing physics-based acoustic simulation to generate realistic chorus recordings by modeling the spatial arrangement, attenuation, and mixing of multiple sound sources.

2.3. Addressing choruses with microphone arrays

A related approach involves using microphone array recordings to perform sound source separation, improving recognition accuracy in chorus conditions (Dai et al., 2021). However, typical blind source separation methods can only separate fewer sound sources than the number of recording channels in the microphone array, making their applicability to insect choruses with dozens of individuals simultaneously chorusing questionable.

2.4. Acoustic simulation tools and physical modeling

Various acoustic simulation tools have been developed to generate realistic soundscapes (Scheibler et al., 2017; Salamon et al., 2017), each adopting different approaches to modeling sound propagation. To simulate cicada choruses with their wide frequency range, we needed a tool capable of accurately modeling physical aspects of sound propagation, including frequency-dependent attenuation due to air absorption.

We selected Pyroomacoustics (Scheibler et al., 2017) as our simulation framework primarily for its excellent capability to model physical sound propagation. Although Pyroomacoustics is designed primarily for indoor acoustics, it can simulate outdoor conditions using the "AnechoicRoom" option, which ignores sound reflection from structures. The main advantage of Pyroomacoustics is its implementation of physics-based propagation modeling, particularly its ability to model

frequency-dependent attenuation where high-frequency sounds attenuate more rapidly with distance than low-frequency sounds. While tools like Scaper (Salamon et al., 2017) simply combine individual sound sources, Pyroomacoustics uses the image source method (Allen and Berkley, 1979) to simulate the path from source to receiver and the air absorption that occurs along this path. This capability is particularly important for simulating natural attenuation of cicada calls that contain high-frequency components. In Pyroomacoustics, the attenuation of sound energy with distance is represented by coefficients dependent on frequency, temperature, and humidity. Further details can be found in the documentation (https://pyroomacoustics.readthedocs.io/en/pypi-release/pyroomacoustics.room.html#air-absorption).

3. Materials and methods

3.1. Overview

In this study, we developed a CNN to recognize the calls of five cicada species commonly found in eastern Japan. The target species were the Large Brown Cicada (*Graptopsaltria nigrofuscata*), the Robust Cicada (*Hyalessa maculaticollis*), the Kaempfer Cicada (*Platypleura kaempferi*), Walker's Cicada (*Meimuna opalifera*), and the Evening Cicada (*Tanna japonensis*). These sympatric species have overlapping adult emergence periods during summer, leading to frequent chorusing. To train an audio recognition model for choruses with minimal effort, we developed a method to artificially create chorus sounds by combining multiple focal recordings obtained from the Internet, a field guide, and our own recordings using directional microphones. This method primarily uses acoustic simulation software (Pyroomacoustics). The CNN model takes spectrograms of audio as input and performs multi-label classification, returning multiple labels. To verify the effectiveness of this method, we varied the number of focal recordings used in each chorus simulation and measured the impact on model accuracy against annotated real cicada choruses. Additionally, we examined optimal settings for chorus simulation using hyperparameter tuning based on performance with real recording data. Fig. 1 shows an overview of the proposed method. The source code is available online (https://github.com/0kam/CicadaChorusDetection).

3.2. Training dataset

As audio sources for model training, we obtained focal recordings of cicada songs from YouTube, audio supplements of a cicada field guide (Masami and Yasumasa, 2015), and new recordings made using directional microphones at our institute. When acquiring cicada recordings from YouTube, we searched using Japanese names for each cicada and selected those focusing on individual cicada songs with minimal noise. The URLs of the acquired sources are available on GitHub (https://github.com/0kam/CicadaChorusDetection/blob/main/step2_prepare_training_sources/youtube_list.md). From these sources, we extracted segments containing only individual cicada song as much as possible and saved them as separate audio files. If a single recording contained multiple songs, they were split into different audio files. This resulted in 8–18 original sources and 18–28 split audio files depending on the species (Table 1). The extracted audio files ranged from 5 to 30 s in length. These files were resampled to 48 kHz, processed with a high-pass filter to remove low-frequency noise (below 500 Hz), and normalized so that the maximum amplitude was 0.8. Fig. 2 shows examples of spectrograms for the target species.

To enhance robustness against noise (such as from other insects and birds), we also used non-cicada sound sources ("other sounds") and background sounds for chorus simulation. For other sounds that might be heard in urban green spaces, we selected bird and insect songs (mainly orthopterans) obtained from YouTube, Xeno-Canto, and insect field guides (Murai and Ito, 2011), as well as environmental sounds from the ESC-50 environmental sound dataset (Piczak, 2015, using all data). For background sounds, we used environmental recordings from three locations at our institute in Ibaraki Prefecture during the summers of 2022 and 2024 (see Section 3.3 for recording details) that did not contain cicada songs (10 ten-minute recordings per site) to account for continuous wind and road noise. These other sounds were processed similarly to cicada songs (resampling, high-pass filtering, normalization), while background sounds were only resampled and normalized.

3.3. Testing dataset

To adjust the training workflow and evaluate the resulting model's performance, we created a field-recorded cicada chorus dataset with expert annotations. This included two datasets: one for tuning and one for testing. These consisted of soundscape recordings made at three locations in our institute during the summers of 2022 (1 location) and 2024 (2 locations). The whole dataset and details about recording locations are available on Zenodo.

For the 2024 recording project, we used ChirpArray (Okamoto and Oguma, 2025) to record for 10 min at the beginning of each hour. Recordings were made at 48 kHz sampling rate with 4 channels and saved as WAV files. For this study, we used only the first channel's recording data. The 2024 recordings were made at two small forest locations in the institute, approximately 200 meters apart. At these two locations, four of the five cicada species were observed, excluding the Evening Cicada.

For the 2022 recording project, we used a custom recording unit consisting of an NVIDIA Jetson Nano 2 GB (NVIDIA Corporation) and a general USB microphone (ELECOM HS-MC09UBK, ELECOM CO.,LTD) to record for 10 min at the beginning of each hour. Recordings were made in mono at 48 kHz and saved in WAV format. This recording site was approximately 100 meters from the 2024 recording sites but consisted of gloomy evergreen forest where Evening Cicadas reside.

From these recording data, we created two annotated datasets: one for tuning and one for testing. To prevent hyperparameter overfitting to the tuning data, we used recordings from different dates and times for the tuning and testing data. Annotation data was created using the following procedure: - Limit to sources recorded between 5:00 and 18:00, from July 15 to September 15 - Randomly select recordings for each recording site (200 recordings per site for testing, 50 recordings per site for tuning) - Extract the first 12 s from each recording - Create labels for cicada species present in each 4 second interval of the 12 second clip (3 clips for each recording)

This procedure allowed us to create annotation data for different audio segment lengths (4, 8, and 12 s). In this study, we used this data to find the optimal audio segment length for model input. This annotation work was performed using Whombat (Balvanera et al., 2025), an open-source acoustic annotation software.

3.4. Chorus simulation

After preparing the training data (cicada songs, other sounds, and background sounds), we simulated cicada choruses using Pyroomacoustics to train the classification model. This software allows simulation of sound reflection, attenuation, and mixing by placing sound sources and microphones in a virtual space. Since our study targeted sounds recorded outdoors, we chose the "AnechoicRoom" virtual space as the simulation environment, which excludes sound reflection from room walls. Using Pyroomacoustics, sound sources and microphones can be placed in 2D or 3D space, and microphone array recordings can be simulated. However, as our study assumes recordings made using a single-channel omnidirectional microphone (e.g., AudioMoth), we conducted the simulation with the microphone and sound sources aligned in a straight line, placing the microphone at the origin (i.e., considering only the distance from the microphone to the sound sources) (Fig. 3). Additionally, as mentioned above, air absorption of sound varies

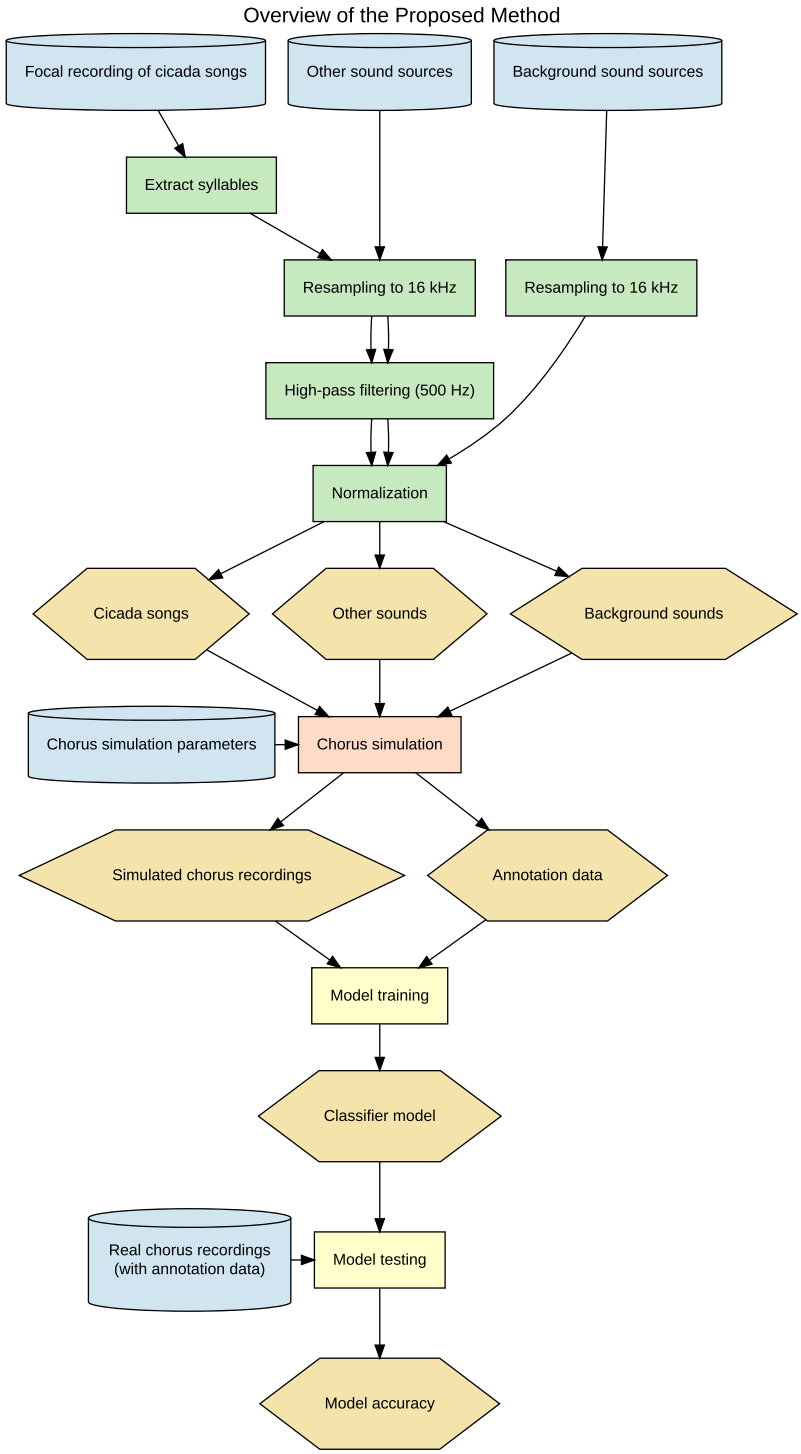

Fig. 1. Overview of the proposed method. This figure illustrates the complete workflow from data collection through preprocessing, chorus simulation, model training, and testing. First, individual cicada calls and environmental sounds are collected from various sources. These are preprocessed and then used as inputs for chorus simulation using Pyroomacoustics, where multiple sound sources are placed at different distances and combinations to mimic real-world chorus conditions. The resulting synthetic choruses are used to train a CNN model. Finally, the model's performance is evaluated against real recording data collected in the field. Components are color-coded by function: cylinders (blue) represent input data sources, boxes represent processing steps (green for audio preprocessing, pink for chorus simulation, yellow for model training/testing), and hexagons represent outputs. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

with temperature and humidity. Since we are targeting recordings in Ibaraki Prefecture during summer, we simulated sound attenuation at 30 degrees Celsius and 70% humidity.

Each simulation included multiple cicada songs, other sounds, and a background sound. For each simulation, we randomly and uniformly selected several parameters from predetermined ranges. These "chorus simulation parameters" included: number of cicada species, number of cicada songs, number of other sounds, types of other sounds (e.g., birds or insects), and the distance from the microphone to each sound source. Table 2 provides examples of these parameter ranges. For each simulation, a background sound was randomly selected and placed 1 meter from the microphone. Each simulation generated a 20 second recording. For sound sources shorter than 20 s, we randomly determined their

Table 1
Number and length of focal recording segments used for training. Each segment contains a single vocalization from an individual with minimal noise when possible. The Evening Cicada has significantly shorter song durations compared to the other four species.

Species	Number of segments	Average segment length (sec)	Total length (sec)
Evening Cicada	28	3.5	98
Robust Cicada	12	49	587
Walker's Cicada	19	22	410
Kaempfer Cicada	22	32	694
Large Brown Cicada	13	40	516

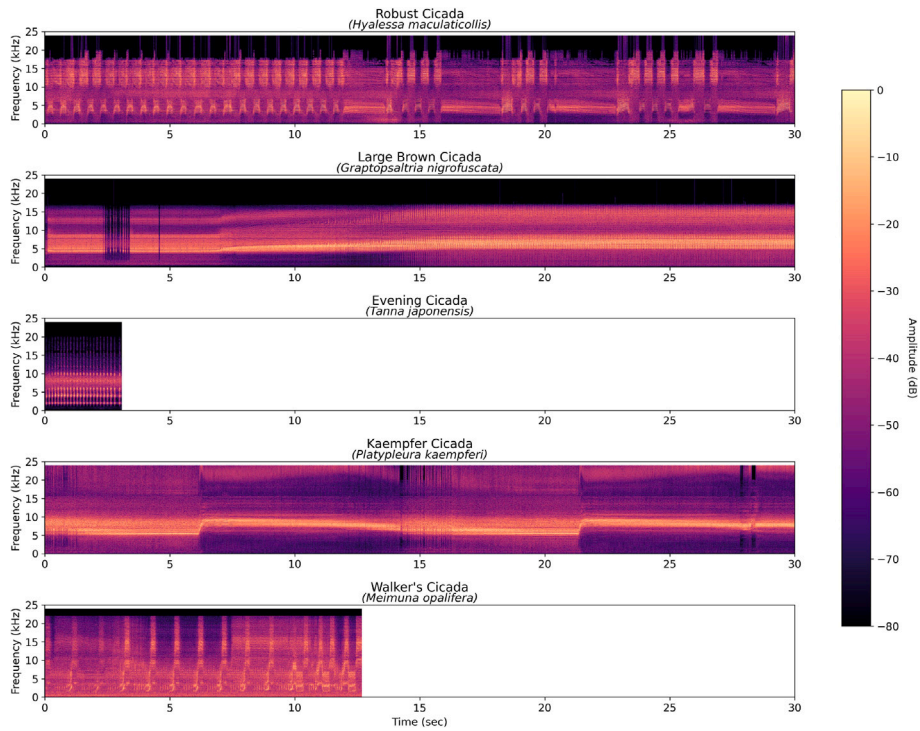

Fig. 2. Examples of each cicada species' songs. The calls of each cicada species, resampled to 48 kHz, were displayed as linear spectrograms. While the calls of each species show distinct patterns, they overlap in frequency. Unlike the other four species, Evening Cicada has syllables that are relatively short, lasting only a few seconds.

playback timing within the simulation period. Longer sources were randomly cut into 20 second segments before playback.

The output of each simulation was a 20 second simulated cicada chorus with an automatically generated annotation file that recorded the start and end times of each cicada song. This approach efficiently produces a large number (thousands) of annotated pseudo-chorus recordings under various conditions with minimal effort. To ensure the model performs well with real choruses, we needed to set the chorus simulation parameters appropriately. We achieved this through hyperparameter tuning using the tuning dataset, as described in Section 3.6.

3.5. Model training

As observed in the real-world dataset (see Section 4.1), the target cicada species frequently form multi-species choruses. Therefore, we trained a multi-label CNN classifier to identify multiple species within a single audio segment. An overview of the training process is shown in Fig. 4 "Model Training". For training the multi-label classification model, we used the sigmoid function as the activation function for the model's final layer and the Mean Squared Error loss as the loss function. To prevent model overfitting, we adopted an early stopping approach. We created a separate validation dataset through simulation, distinct from our training data. During each training epoch, we evaluated model performance on this validation data and selected the model from the epoch that showed the best validation performance.

3.6. Hyperparameter tuning

We expected model performance to vary depending on model architecture, input specifications, and chorus simulation parameters (e.g., the number of cicada songs to include at once). We optimized these variables to maximize model accuracy on the tuning dataset. For both tuning and testing with real-world recording data, we used the average F1 score (Eq. (1), Chinchor and Sundheim, 1993) across all species as our accuracy indicator.

$$\begin{aligned} precision &= \frac{TP}{TP+FP} \\ recall &= \frac{TP}{TP+FN} \\ F_1 &= \frac{2 \cdot precision \cdot recall}{precision + recall} \end{aligned} \tag{1}$$

3.6.1. Model input and architecture tuning

First, we determined the length of audio segments the model would receive as input, the sampling rate, and the model architecture. We searched for the optimal combination from five representative CNN models commonly used for image classification and audio recognition tasks (ResNet50 (He et al., 2016), EfficientNetB0, EfficientNetB4 (Tan and Le, 2019), EfficientNetV2S, EfficientNetV2M (Tan and Le, 2021)), three audio segment lengths (4, 8, 12 s), and four sampling rates (16 kHz, 24 kHz, 36 kHz, 48 kHz). Chorus simulations were performed

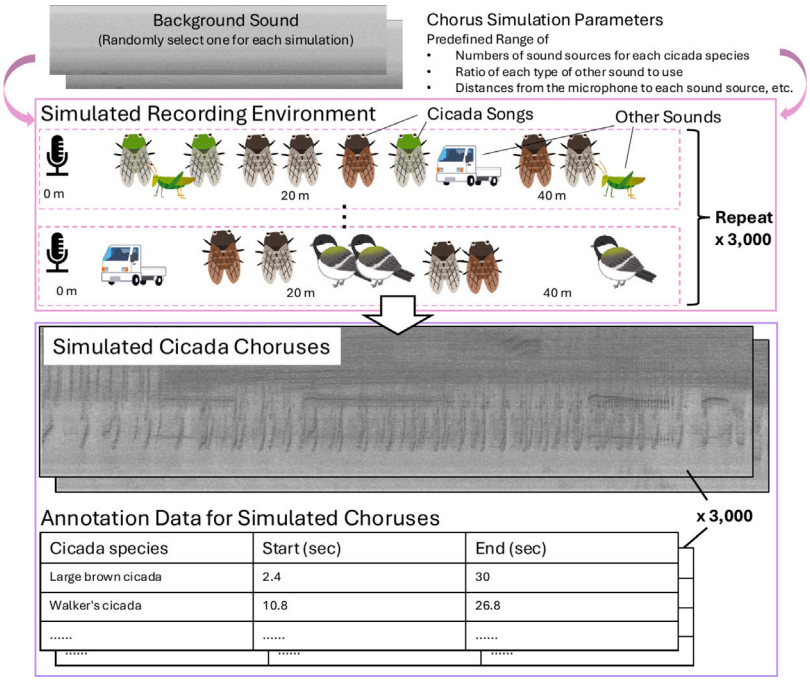

Fig. 3. Overview of the proposed chorus simulation method. In each simulation, we randomly selected cicada songs, other sounds, and a background sound and placed them using a randomly selected simulation settings (e.g., number of cicada songs, distance from the microphone to each sound sources). The range of each settings (e.g., max / min of the number of cicada songs) were predefined as chorus simulation parameters. The output is a 20 s of simulated cicada chorus with annotations that mark the start and end of each cicada song. To exhaustively simulate the various choral patterns, the process was repeated 2000 to 3000 times (depending on the experiment).

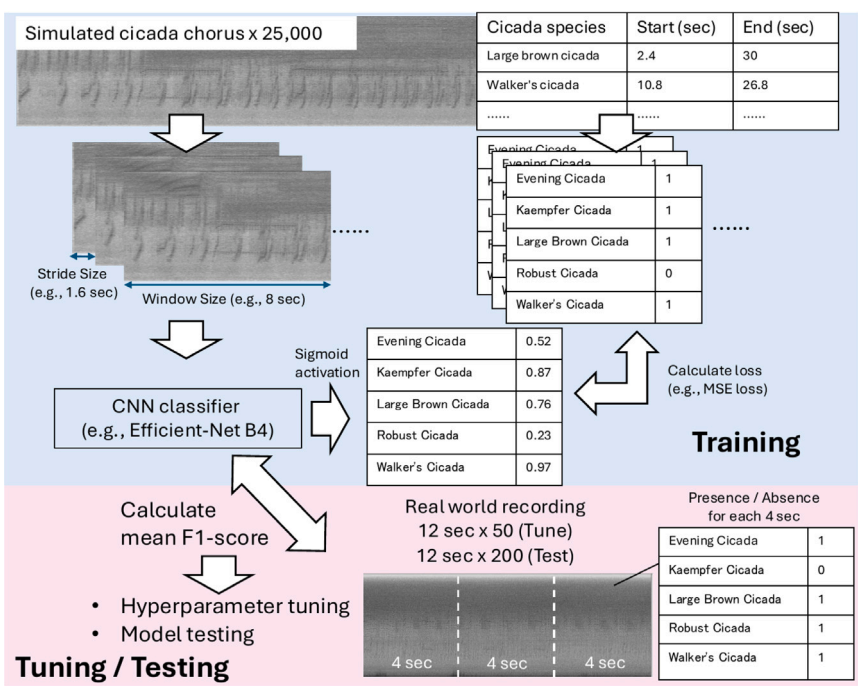

Fig. 4. Overview of the model training and testing. We trained a multi-label CNN classifier that takes an eight-second audio segment and outputs a multi-hot label indicating the presence or absence of each cicada species. The trained model was then tested on real soundscape recordings. For testing, 12 second audio segments were annotated at 4 second intervals to create testing and tuning data at different temporal resolutions.

2000 times for training data and 500 times for validation data, and model training was conducted with a batch size of 6 for 20 epochs.

We also optimized volume adjustment to account for differences in recording conditions between simulation and real environments. In Pyroomacoustics, each sound source pre-processed to have a maximum amplitude of 0.8 (see Section 3.2) is subject to air absorption based on temperature and humidity, and is recorded at its resulting amplitude. In contrast, actual field recordings have smaller signal amplitudes compared to the simulation environment due to microphone sensitivity and recording equipment gain settings.

Therefore, we tested multiple volume amplification values (+10 dB, +15 dB, +20 dB, +25 dB, +30 dB) on the tuning dataset and evaluated the model's detection performance. As a result, +15 dB amplification was selected and then we applied this value to all real field recordings.

For other hyperparameters, please refer to the experimental configuration file (https://github.com/0kam/CicadaChorusDetection/blob/main/step3_model_training/configs/tune_model.yaml).

3.6.2. Chorus simulation tuning

Next, we optimized chorus simulation parameters. Specifically, we adjusted the proportion of data containing cicada songs in the simulation, the range of the number of cicadas and other sound sources to include in each simulation, the range of distances from sound sources to the microphone in the simulation environment, and which types of other sound sources (bird songs, non-cicada insect songs, ESC-50 dataset) to prioritize. Due to the large number of chorus-related parameters, validating all combinations was computationally challenging. Therefore, we used the Tree-structured Parzen Estimator sampler implemented in Optuna (Ozaki et al., 2022) and conducted 100 trials. Chorus simulations were performed 3000 times for training and 500 times for validation, and model training was conducted with a batch size of 8 for 20 epochs. For hyperparameter search conditions and other hyperparameters, please refer to the experimental configuration file (https://github.com/0kam/CicadaChorusDetection/blob/main/step3_model_training/configs/tune_simulation_tpe.yaml).

3.6.3. Evaluating the impact of maximum cicada individuals in chorus simulation

In this study, we hypothesized that models trained on simple focal recordings would be insufficient to handle complex real-world cicada choruses, and that this limitation could be overcome through soundscape generation via acoustic simulation. To test this hypothesis, we investigated how the performance of models trained on the resulting data changed on the testing dataset as we varied the maximum number of cicada sound sources included in chorus simulations. Specifically, we fixed the minimum number of cicada sources at 1 and varied the maximum number as 1, 2, 5, 10, 20, and 40. We used the best values obtained from the above tuning for other parameters.

3.7. Model testing

Finally, we evaluated performance on the testing dataset using the best model obtained through chorus simulation optimization.

3.8. Application to long-term soundscape recordings

To demonstrate the potential application of the trained model, we selected one of the 2024 recording sites used for the testing/tuning dataset and applied the model to the entire soundscape recordings collected from June to September 2024. This application example shows how the model can be used to analyze temporal patterns of cicada activity, although detailed verification of these patterns was beyond the scope of this study.

4. Results

4.1. Cicada species composition in real-world recording datasets

Fig. 5 shows the number of occurrences and combinations of cicada species in the tuning and testing datasets. Here, species were counted for each 4 second audio segment. The Evening Cicada, which is found only at one recording site and has limited singing time, had fewer recordings compared to the other four species, but approximately equal numbers of recordings were obtained for the other four species. Of the entire dataset, 11.6% contained no cicada songs, 44.1% contained 1 species, 35.1% contained 2 species, 8.7% contained 3 species, and 0.5% contained 4 species. Since these cicada songs (except for the Evening Cicada) are longer than 4 s (see Table 1), this indicates that approximately 40% of the recordings feature two or more cicada species chorusing simultaneously.

4.2. Model input and architecture tuning results

The combination of 4 second segments, 16 kHz sampling rate, and EfficientNetB4 architecture was selected for the model input. Although model architecture did not have a significant impact on performance, there was a general trend of decreasing performance with increasing sampling rate (Fig. 6).

4.3. Chorus simulation hyperparameter tuning results

Based on the optimized chorus simulation parameters (Table 2), including exactly 5 cicada songs in every simulation was selected (cicada_n_min = cicada_n_max = 5). This indicates that model performance improves when consistently learning from simulations with multi-individual choruses. The mean F1 score for the model trained on chorus data simulated using the optimized parameters was 0.83 (Table 3).

4.4. Impact of maximum cicada individuals in chorus simulation

Fig. 7 shows the relationship between the maximum number of cicada songs in each simulation (cicada_n_max) and the mean F1 score on the testing dataset. Using only one focal recording generated a mean F1 score of 0.59; in contrast, using at most 20 cicada songs improved it by 22% (i.e., 0.22 increase), to 0.81. However, increasing cicada_n_max to 40 slightly reduced model performance. Note that in the simulation parameter optimization, both the minimum and maximum number of cicada songs were set to 5. This suggests that in this experiment, where the minimum number of cicada songs was fixed at 1, a larger maximum number was selected to increase training data containing choruses.

4.5. Application to long-term soundscape recordings

Fig. 8 visualizes the temporal patterns of cicada song activity detected when the model was applied to the entire soundscape recordings. In the circular graphs, the angle represents time of day, the distance from the center represents date, and the opacity corresponds to the frequency of songs detected within each ten minutes recording in each hour. This visualization suggests potential differences in activity patterns between species, such as Kaempfer Cicada emerging seasonally earlier compared to other species and Large Brown Cicada starting singing relatively late in the morning. These patterns should be interpreted as preliminary observations and require independent ecological verification, which was not conducted in this study.

5. Discussion

The mismatch between acoustic domains (focal vs. soundscape) in training and testing datasets has presented a major bottleneck in applying machine learning to soundscape recordings (Kahl et al., 2021), especially for organisms such as cicadas, which produce large and complex choruses. Our approach bridges this gap by simulating realistic soundscapes using multiple focal recordings. This significantly improved model accuracy, allowing for accurate detection of multiple cicada species in noisy choruses with just tens of focal recordings per species. Compared to previous data augmentation methods like MixUp, our method enables training of acoustic classification models with more complex and realistic sound sources incorporating frequency-dependent sound distance attenuation and the overlapping of sounds from multiple individuals. In fact, increasing the maximum number of cicada sound sources used in each simulation from 2 to 20 resulted in substantial performance improvement. Unlike previous studies involving blind audio-source separation techniques (Dai et al., 2021), our approach does not require specialized hardware such as microphone arrays. Additionally, compared to approaches using deep generative models (Herbst et al., 2024), our approach offers advantages in terms of lower computational cost and easier determination of simulation parameters based on ecological knowledge of the target species.

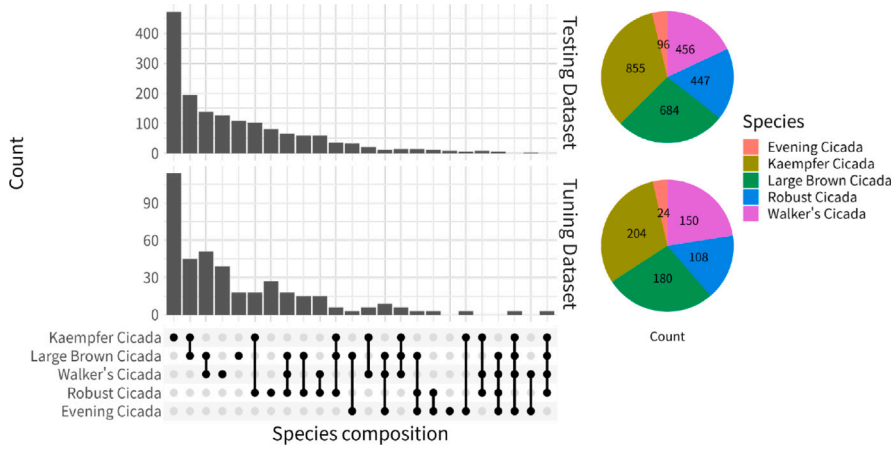

Fig. 5. Number of occurrences and combinations of cicada species in the tuning and testing datasets. The pie charts show the count of 4 second audio segments containing each cicada species, while the vertical bars represent the count of different species combinations. Each 4 second segment was individually annotated for the presence/absence of each species. The Evening Cicada appears less frequently due to its limited geographical distribution within our recording sites. Approximately 40% of the recordings feature two or more cicada species chorusing simultaneously.

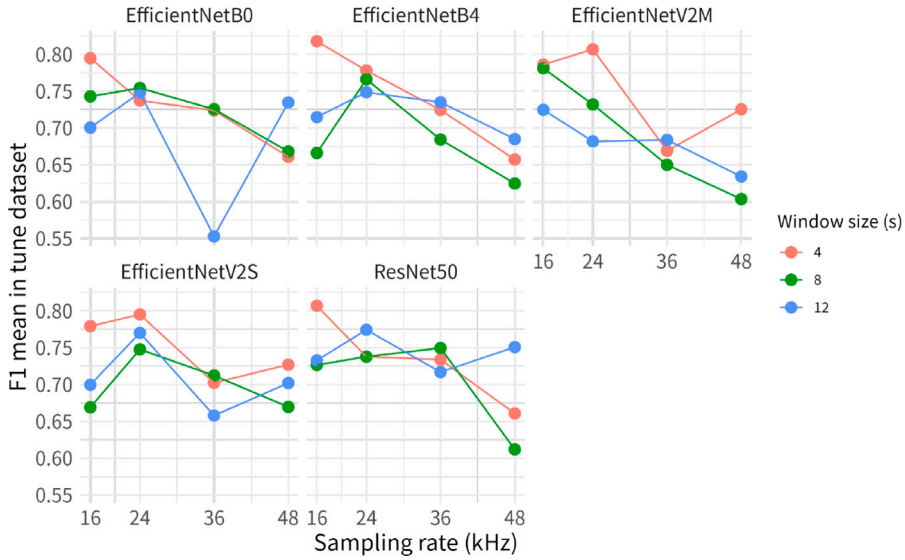

Fig. 6. Performance comparison by model input and architecture. The plot shows mean F1 scores on the tuning dataset for different combinations of CNN architectures (EfficientNetB0, EfficientNetB4, EfficientNetV2M, EfficientNetV2S, and ResNet50), sampling rates (16, 24, 36, and 48 kHz), and audio window sizes (4, 8, and 12 s). The combination of EfficientNetB4 architecture, 16 kHz sampling rate, and 4 second window size achieved the highest performance. A general trend of decreasing performance with increasing sampling rate is observed across most architectures and window sizes.

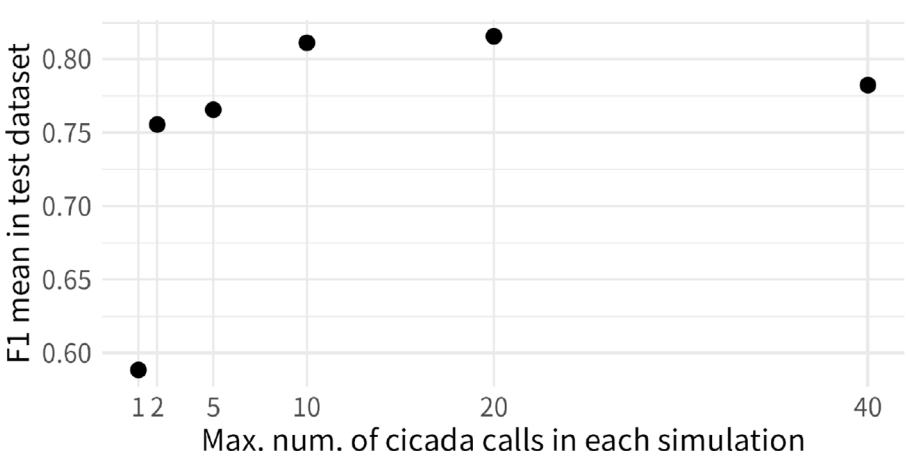

Fig. 7. Effect of chorus simulation on model performance. The X-axis shows the maximum number of cicada songs in each simulation (cicada_n_max), and the Y-axis shows the mean F1 score for each species. In each simulation, the number of cicada songs was randomly selected from 1 to cicada_n_max. The model performed best against the testing dataset at a cicada_n_max value of 20.

Table 2
Optimized chorus simulation parameters. For each simulation, parameters such as the number of cicada songs, number of other sounds, and distances between sound sources and the virtual microphone were randomly selected within these optimized ranges. These parameters were determined through hyperparameter tuning to maximize model performance on the tuning dataset.

Parameter name	Overview	Optimized parameter
cicada_n_min	Minimum number of cicada songs in each simulation	5
cicada_n_max	Maximum number of cicada songs in each simulation	5
cicada_distance_min	Minimum distance from the microphone to each cicada song (m)	60
cicada_distance_max	Maximum distance from the microphone to each cicada song (m)	120
others_ratio	Ratio of simulations lacking cicada songs	0.1
others_n_min	Minimum number of other sounds in each simulation	0
others_n_max	Maximum number of other sounds in each simulation	5
others_distance_min	Minimum distance from the microphone to each other sound (m)	10
others_distance_max	Maximum distance from the microphone to each other sound (m)	120
others_weights	Weights of different types of other sounds (bird, insect, esc50) used in the simulation	bird: 1.0, esc50: 1.0, insect: 1.0

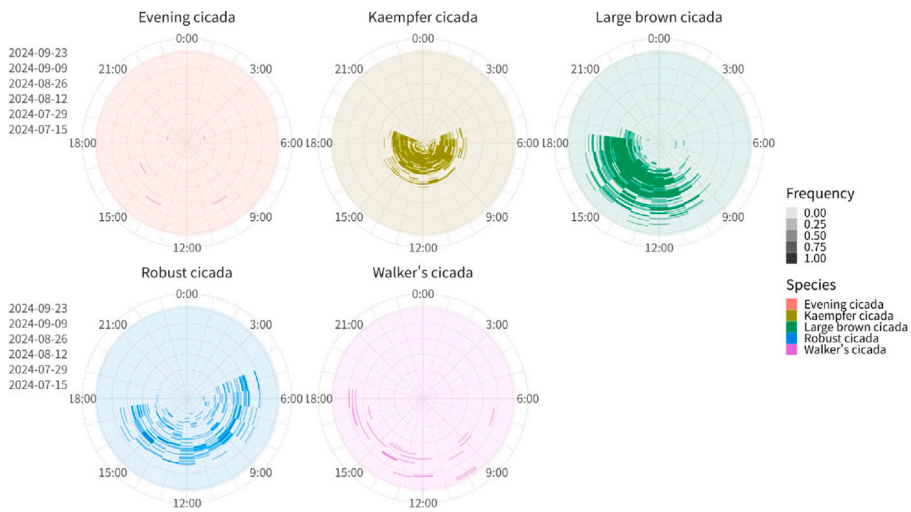

Fig. 8. Temporal patterns of cicada singing activity. The visualization shows cicada activity patterns detected by applying our model to continuous soundscape recordings collected at our research site in Ibaraki, Japan. Each circular plot represents one cicada species, with time of day shown as the angle (starting from midnight at the top, proceeding clockwise) and date as the distance from the center (June to September 2024). Color opacity indicates the frequency of detected songs within a ten-minutes recording in each hour. Notable patterns include the early seasonal emergence of Kaempfer Cicada compared to other species and the Large Brown Cicada's tendency to begin singing relatively late in the morning.

Table 3
Performance of the optimal model. Using the optimized chorus simulation parameters, our approach enabled accurate classification on field recordings. Walker's Cicada shows comparatively lower recall (0.58), likely because its song contains characteristic patterns in both easily attenuated high-frequency ranges (around 15 kHz) and low-frequency ranges (around 1.5 kHz) that often overlap with other environmental sounds, making detection more challenging in chorus conditions.

Species	F1-score	Precision	Recall
Evening Cicada	0.81	0.76	0.86
Kaempfer Cicada	0.88	0.80	0.97
Large Brown Cicada	0.92	0.90	0.93
Robust Cicada	0.85	0.83	0.86
Walker's Cicada	0.71	0.91	0.58

5.1. Hyperparameter tuning

The optimal input parameters identified through tuning were unexpected. Despite cicada songs containing high frequencies (above 24 kHz), a 16 kHz sampling rate (capturing up to 8 kHz on spectrograms) performed best. This counterintuitive result may relate to spectrogram resolution constraints. CNN architectures require fixed input dimensions (380 × 380 pixels for EfficientNetB4), and higher sampling rates may reduce the visibility of important spectral features. Similarly, shorter audio segments (4 s) were preferred, likely for the same reason. Future improvements could involve separate processing of different frequency bands before integration, adoption of models supporting higher resolution inputs, or exploration of alternative architectures such as Vision Transformers or Recurrent Neural Networks that might better capture detailed patterns in frequency domain.

The chorus simulation parameters confirmed the importance of multi-individual simulations for model accuracy. Our experiments showed that including up to 20 sound sources in each simulation maximized performance, with a notable decrease in accuracy when using only single-source recordings. During optimization, both minimum and maximum cicada sources converged to 5, indicating the importance of consistently including multiple individuals. Given that four of the five target species produce songs longer than our 20 second simulation window (see Table 1), these simulations effectively represent simultaneous chorusing by multiple individuals.

5.2. Contributions and limitations

5.2.1. Contributions

The primary contribution of this study is the development of an acoustic simulation-based approach for cicada song recognition in complex chorus conditions. Our method demonstrates that highly accurate models can be built with limited focal recordings, significantly reducing the manual annotation effort traditionally required for ecological monitoring. This innovation is particularly valuable for research involving chorusing species, where conventional data collection methods are exceptionally labor-intensive. Though we focused on cicadas, this approach is readily adaptable to other chorusing taxa such as frogs and crickets. Further research should evaluate its effectiveness across diverse taxonomic groups and ecosystems.

5.2.2. Limitations

Several limitations of our study should be acknowledged. First, our evaluation was conducted at a single research site within a small area (locations several hundred meters apart). The model's performance in different environments – such as urban areas with distinct noise profiles or varied vegetation types – remains unverified. Additionally, our hyperparameter tuning used recording data from the same sites as our testing data, which may limit generalizability. A conservative interpretation of our results is that the model performs well on audio from the same recording site used for tuning. While this scope may seem limited, it still substantially reduces annotation effort compared to conventional training methods. This suggests our approach may be more suitable for fixed-position ARUs than for diverse citizen science data collected via smartphone apps. Future research should compare accuracy across recording sites with different environmental characteristics.

We currently use annotated data to evaluate simulation fidelity, but this could potentially be replaced with annotation-free methods. For example, acoustic autoencoders (Rowe et al., 2021) could compare characteristics between real and simulated soundscapes to evaluate realism and optimize parameters.

Our simulations using Pyroomacoustics – originally designed for indoor acoustics – do not fully capture outdoor acoustic phenomena such as wind effects or vegetation-based attenuation. Incorporating these factors could further improve performance.

When applying our method to other taxa (e.g., orthopterans, amphibians) or geographical regions, simulation parameters would need adjustment to match the acoustic characteristics of those environments and organisms. In environments with higher species diversity than our five-species system, modifications to model architecture and training strategies may be necessary.

Despite these limitations, our study presents a novel approach to species identification in complex acoustic environments, contributing to more efficient ecological monitoring. Future work will extend this approach to orthopterans and amphibians and verify its generalization capability across diverse environmental conditions and taxa.

5.3. Potential applications

Our preliminary application of the model to continuous recordings (Fig. 8) demonstrates its potential for monitoring temporal and seasonal patterns of cicada activity. While we did not conduct detailed ecological verification, this visualization shows how simulation-trained models can generate hypotheses about species-specific phenology and diurnal activity cycles. Future research could correlate these patterns with independent ecological observations and extend to multi-year monitoring to track changes in emergence timing related to climate change. For example, Tsujimoto et al. (2024) reported that higher temperatures are advancing the first singing date of the Large Brown Cicada (*Graptopsaltria nigrofuscata*), one of our target species. Our method could provide an automated tool for large-scale monitoring of such climate-driven phenological shifts across multiple species simultaneously.

This approach also has implications for acoustic community research (Chhaya et al., 2021), particularly studying interspecies interactions through vocalizations. In multi-species chorusing environments, acoustic masking commonly occurs (Schmidt and Balakrishnan, 2015). Our method, which accurately detects species composition within choruses, provides a valuable tool for investigating the extent and patterns of interspecies acoustic masking in field conditions. For such research, combining our acoustic simulation approach with deep learning-based sound source separation methods like MixIT (Wisdom et al., 2020) could enable estimation of not only species presence/absence but also the number of vocalizing individuals (at least for species with non-stationary songs). This would provide more comprehensive insights into acoustic masking phenomena and community dynamics.

6. Conclusion

In this study, we presented a simulation-based approach for enhancing automatic detection of cicada songs in complex chorus conditions. By generating thousands of simulated chorus recordings through physical acoustic modeling, we demonstrated the effectiveness of our method in accurately identifying multiple cicada species in complex soundscapes, achieving a mean F1 score of 0.83 on real-world recordings—a 22% improvement over models trained solely on focal recordings.

The ability to train high-performance models with minimal labeled data (approximately 10 focal recordings per species) significantly reduces annotation burden while maintaining detection accuracy, making systematic acoustic monitoring more feasible for ecological applications. Our method demonstrates that realistic acoustic simulation can effectively bridge the domain gap between focal recordings and complex soundscapes.

While we focused on cicadas in this study, the methodology is applicable to other acoustically active taxa, providing a scalable solution for biodiversity monitoring in complex acoustic environments. As passive acoustic monitoring continues to expand as a non-invasive survey technique, approaches like ours will play an important role in understanding and conserving biodiversity in a rapidly changing world.

CRediT authorship contribution statement

Ryotaro Okamoto: Writing – review & editing, Writing – original draft, Software, Methodology, Data curation, Conceptualization. **Hiroyuki Oguma:** Writing – review & editing, Data curation, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The source code is available online at Github. Also, annotated soundscape recordings used in the testing and tuning data is available at Zenodo.

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